

Gumbel (Type I) Extreme Value Distribution

(Tool 40)
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RESULT

0.9900 (dimensionless)

OUTPUTS

OUTPUT	VALUE
Return period $T = 1/(1-F)$ of the supplied x	100.0000 years
Gumbel reduced variate $y = -\ln(-\ln F)$	4.6001 (dimensionless)
100-year return level $\mu - \beta \ln(-\ln(1-1/100))$ (input units)	192.0030 (input units)

FIGURE 1 · PROBABILITY DENSITY FUNCTION

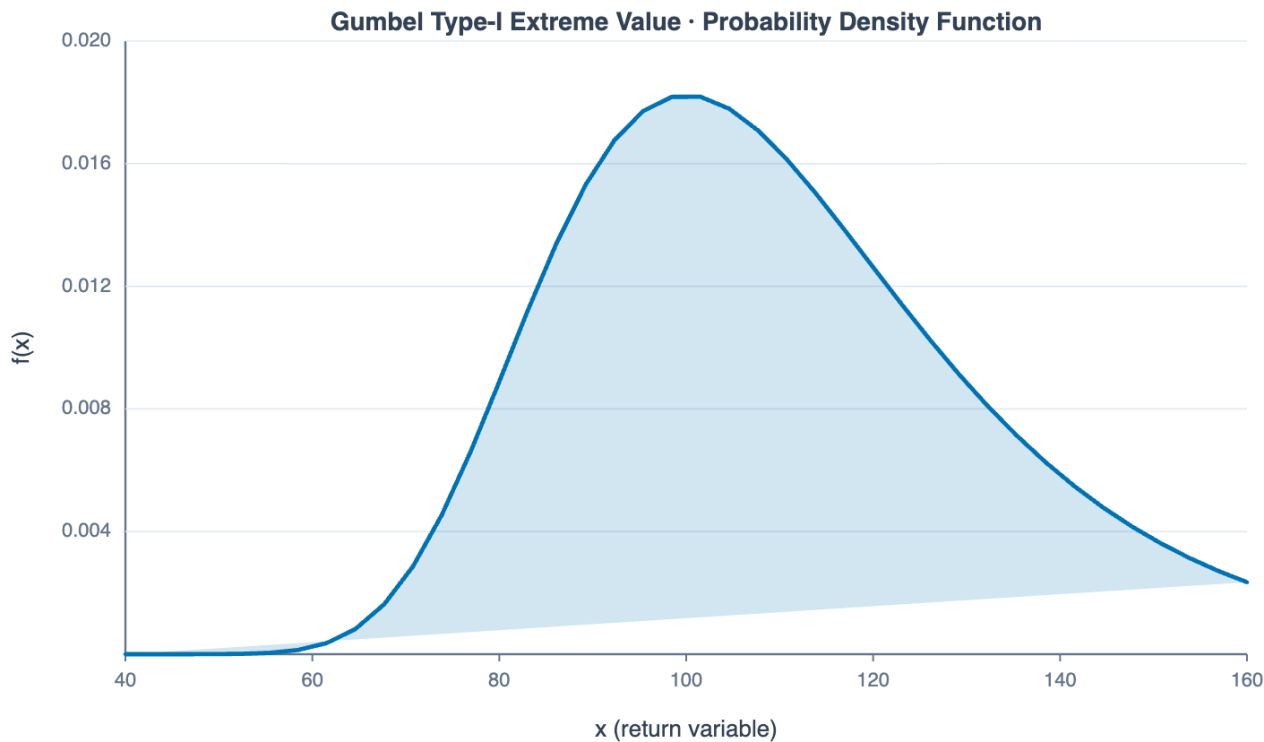


Figure 1 — Gumbel PDF $f(x)$ for $\mu = 100$, $\beta = 20$ (return-period filter = 100 yr).

DATA SOURCES

open-meteo-weather, open-meteo-marine, open-meteo-aq, srtm-elevation, noaa-space-weather, era5-proxy-friction-velocity, era5-tcwg, era5-proxy-surface-fluxes, era5-soil-state, ghcn-climate-normals-r-factor

INTERPRETATION

Gumbel Extreme Value Analysis: 0.9900. Gumbel distribution: $F(x) = \exp(-\exp(-(x-\mu)/\beta))$. Return period: $x_T = \mu - \beta \ln(-\ln(1-1/T))$.

RECOMMENDATIONS

- 01 Use ≥ 30 years of data.
- 02 Compare Gumbel and GEV via AIC/BIC.
- 03 Cross-reference USGS stream gauges for floods.

METHODOLOGY

Location $\mu = 100.00$, Scale $\beta = 20.00$

Value $x = 192.00$

Step 1 — Reduced variate:

$$y = (x - \mu) / \beta = (192.00 - 100.00) / 20.00 = 4.6001$$

Step 2 — CDF:

$$F(x) = \exp(-\exp(-y)) = \exp(-\exp(-4.6001))$$

$$P(X \leq 192.00) = 0.990000$$

Step 3 — Return period:

$$T = 1 / (1 - F) = 1 / (1 - 0.990000) = 100.0 \text{ years}$$

Step 4 — Gumbel plot slope check:

Gumbel reduced variate: $y = 4.6001$

· Gumbel plot of $\ln(-\ln(F))$ vs x should be linear if data follow Type I

- Return period: Rare (50-100 yr)

— End of report · Generated by Terranoetis —